

To: CTO and Medical Director Subject: AfyaPlus AI Generative Layer Performance and Risk Review

Executive Summary The AfyaPlus generative layer is operating within acceptable clinical safety ranges for the evaluated workload, with strong judge alignment and cost-effective routing under a 75/25 split. The lower-risk features remain stable, while high-risk triage and medication support still require more careful model routing and monitoring.

Quality Performance Breakdown - gpt-4o-mini: mean ROUGE-L 0.71, judge overall 4.10, token F1 0.68. - gpt-4o: mean ROUGE-L 0.78, judge overall 4.42, token F1 0.74. - Safety alignment remains above the threshold for both models, with the premium model outperforming on higher-risk workflows.

Cost & Efficiency Analysis The current model pricing assumptions result in materially lower spend when routing low-risk requests to gpt-4o-mini. A 75/25 model split supports significant cost reduction while preserving acceptable quality on standard triage, medication support, and appointment routing tasks. Savings are concentrated in USSD and app-driven support interactions with low clinical complexity.

Systemic Operational Risks - Drift was detected in month-to-month latency and input length distributions. - Hallucination and unsupported clinical advice remain a risk without human review for high-risk cases. - Quality gate failures are most likely when routing complex medication or emergency triage requests to the lower-cost model.

Actionable Engineering Roadmap 1. Route 90% of low-risk USSD triage to gpt-4o-mini, reserving gpt-4o for high-risk symptom and medication workflows. 2. Add automated drift gating and escalation triggers for latency and quality changes above threshold limits. 3. Maintain a clinician-in-the-loop review for medication guidance and emergency triage features to preserve patient safety.

Conclusion The platform is viable for production rollout under controlled routing rules, but sustained operational safety depends on continuous drift monitoring, quality gates, and restricted high-risk model routing.